

Knowledge-Aware Evaluation for Legal Document Summarization

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Abstract—Traditional methods such as BLEU and ROUGE evaluate summaries mainly through surface-level text similarity, without checking whether the legal summary actually preserves the original intent and meaning of the case. This limitation was previously addressed by Mullick et al. [9] through the Intent Metric, which uses exact substring matching between annotated phrases and summary sentences. However, exact matching fails when summaries are paraphrased versions of the original phrases. In this paper, we introduce two improved approaches: (1) a Semantic Intent Metric (SIM), which measures semantic similarity using MPNet sentence embeddings with windowed cosine similarity, and (2) an LLM-as-Judge method, which evaluates intent preservation using a structured prompting framework. Experiments were conducted on 93 Indian legal documents across four categories of cases. Results show that only SIM demonstrates a statistically significant positive Spearman rank correlation with the LLM-as-Judge approach ($\rho = +0.255$, $p = 0.014$). In contrast, BLEU, ROUGE-L, and the exact-match Intent Metric show weak negative correlations. The proposed SIM performs better for legal intent paraphrasing tasks such as Corruption and Land Dispute cases, but is comparatively less effective for Murder and Robbery cases that rely heavily on physical descriptions.

Index Terms—legal summarization, evaluation metrics, intent extraction, semantic similarity, LLM-as-Judge

I. INTRODUCTION

A lawyer who is practising law often has to read hundreds of pages of documents about a case to find the facts. The decisions made by courts are usually very long. For example, a decision made by the Indian Supreme Court can be around 4,500 words [9]. Summarizing things automatically can help make the documents shorter, but it is only helpful if the summary still has the important legal parts of the case. These parts include phrases like “*preparation to kill*”, “*adverse possession*”, or “*accepted illegal gratification*”. These phrases show what someone was thinking when they did something in the case. We call these phrases *intent phrases* because they reflect the intent behind actions and are very important in law.

We have a problem with how we measure if a summary still has the idea of the original text. Some metrics, like BLEU [10] and ROUGE [6] just check how many words are the same in the text and the summary. They do not think about how important each word or phrase is.

Other metrics like BERTScore [17] look at how similar the meanings are. They also do not focus on the specific phrases that make a case what it is.

Mullick and others created the Intent Metric [9]. This metric checks how well a summary matches the phrases in the text. It looks at the phrases that show what the text is about. It checks if they are in the summary.

The Intent Metric agrees more with what people think. This was shown in tests on 101 documents and 59 Australian legal documents. The Intent Metric matched what people thought better. It did this better than BLEU or ROUGE.

The Intent Metric is good. It measures if a summary still has the idea of the original text. It does this by focusing on the phrases that make a case what it is. The Intent Metric is better than BLEU, ROUGE, and BERTScore.

The original Intent Metric has a structural limitation: substring matching is brittle to paraphrase. A summary that captures the meaning of “*preparation to kill*” as “*planned the killing*” receives zero credit. In legal summaries this is common: extractive summarizers select sentences whose wording differs from the annotated phrases, and abstractive summarizers can rephrase aggressively. Validating an evaluation framework against actual human ratings, as Mullick et al. did using crowd-sourced annotation, is also expensive and difficult to reproduce.

We make two contributions:

- 1) We propose a **Semantic Intent Metric (SIM)** that replaces the binary substring match of [9] with cosine similarity over MPNet [16] sentence embeddings, computed over sliding word-windows inside each summary sentence. SIM identifies semantic equivalents of intent phrases that the original metric misses while preserving the same precision/recall/F1 framework.
- 2) We integrate an **LLM-as-Judge** component [7], [18] that prompts a small open-source language model to score intent preservation on a 1–5 Likert scale. This provides a reproducible, zero-cost reference signal that does not require crowd-sourced human ratings.

We evaluate both contributions on the Indian-Data corpus released by [9]: 93 documents annotated with intent phrases across four legal categories. SIM is the only automated metric that correlates positively and significantly with the LLM-as-Judge reference. The original Intent Metric, BLEU, and ROUGE-L all correlate negatively, indicating that they reward verbatim copying which does not reliably translate to preservation of legal intent.

II. RELATED WORK

A. Legal Document Summarization

Legal document summarization methods range from rule-based extractive systems such as LetSum [5] and CaseSummarizer [11], to graphical-model approaches that apply CRFs over rhetorical roles [15], and transformer-based extractive [8] as well as abstractive models [1]. Bhattacharya et al. [2] presented a comparative analysis of Indian Supreme Court judgments. Domain-specific models like Legal-BERT [3] have improved performance in legal classification tasks, but they were not specifically developed for evaluating legal summarization quality.

B. Summarization Evaluation

Natural Language Generation (NLG) evaluation methods have been widely studied by Sai et al. [14]. Traditional lexical metrics such as BLEU, ROUGE, and METEOR mainly measure overlap between words and phrases in generated and reference texts. In contrast, embedding-based metrics like BERTScore, MoverScore, and SMS [4] evaluate semantic similarity using contextual embeddings. However, these methods are not specifically designed for the legal domain, as they treat all tokens with equal importance and do not account for legally significant terms or intent. To address this limitation, Mullick et al. [9] introduced the first intent-aware evaluation metric for legal summarization, which directly influenced the development of our proposed approach.

C. LLM-as-Judge

Researchers have increasingly used Large Language Models (LLMs) for automated evaluation tasks. For example, G-Eval [7] uses GPT-4 with a structured set of evaluation criteria to assess generated outputs. Zheng et al. [18] proposed the idea of using language models as judges and studied the potential biases associated with this approach. Inspired by these methods, we apply the LLM-as-Judge framework to legal summarization evaluation. Our approach provides an open-source language model with the original legal document, a generated summary, and intent-related phrases to determine how effectively the summary preserves the important legal meaning of the case.

III. METHODOLOGY

A. Original Intent Metric

We begin by revisiting the metric proposed by Mullick et al. [9]. Consider a legal document containing a set of intent-related phrases represented as P_i , where the total number of phrases is M . Let the generated summary consist of N sentences, represented as O_j . The objective is to measure how effectively the summary preserves the intent expressed in the original document. To achieve this, a similarity function is defined to compare each intent phrase from the document with the sentences present in the summary.

$$s_{ij}^{\text{orig}} = \mathbb{1}[P_i \text{ is a substring of } O_j]. \quad (1)$$

Precision, recall, and F1 are computed as:

$$P = \frac{1}{N} \sum_{j=1}^N \mathbb{1}[\sum_i s_{ij} > 0], \quad (2)$$

$$R = \frac{1}{M} \sum_{i=1}^M \mathbb{1}[\sum_j s_{ij} > 0], \quad (3)$$

$$F_1 = \frac{2PR}{P+R}. \quad (4)$$

B. Semantic Intent Metric (SIM)

We replace the substring-matching approach with semantic similarity. Let $e(\cdot)$ denote a sentence-tuned embedding model, where we use `all-mpnet-base-v2` [?], [16]. For each summary sentence O_j , we generate sliding word windows $W_j^{(L)}$ of length L , approximately equal to the corresponding phrase length, with 50% overlap. The semantic similarity score is defined as:

$$s_{ij}^{\text{SIM}} = \mathbb{1}\left[\max_{w \in W_j^{(L_{P_i})}} \cos(e(P_i), e(w)) \geq \tau\right]. \quad (5)$$

Two design choices are particularly important in our approach. First, comparing a short intent phrase with an entire long summary sentence can reduce the cosine similarity score because large parts of the sentence may contain unrelated context. To address this issue, we compare phrases against length-matched sliding windows, which helps localize the semantic comparison more effectively. Second, we use sentence-tuned MPNet embeddings instead of raw Legal-BERT embeddings because, in our pilot experiments, mean-pooled token embeddings from Legal-BERT produced representations that were too closely clustered to clearly distinguish between matching and non-matching phrases. The threshold τ is selected through a threshold sweep procedure described in Section IV-C.

C. LLM-as-Judge

We use a Large Language Model (LLM) to evaluate how well a generated summary preserves the meaning and intent of the original legal document. The model is provided with the source document, a candidate summary, and a set of annotated intent phrases, and is asked to assign a score on a 1–5 Likert scale, where 1 represents very poor intent preservation and 5 represents excellent preservation. The prompt is designed to produce a structured JSON response containing both a numeric score and a one-sentence justification, which can be parsed automatically. For our experiments, we use `Qwen2.5-1.5B-Instruct` [12] as a local, open-source, and reproducible backend. The framework also supports API-based evaluation using models such as Anthropic Claude and OpenAI GPT-4.

D. Evaluation Protocol

We evaluate three intent-aware metrics discussed earlier: the Original Intent Metric, the Semantic Intent Metric (SIM), and the LLM-as-Judge approach. In addition, we include two commonly used lexical evaluation metrics, BLEU and

TABLE I
INDIAN-DATA STATISTICS. AVG. PHRASES IS THE AVERAGE NUMBER OF ANNOTATED INTENT PHRASES PER DOCUMENT.

Category	# docs	avg. words	avg. sents	avg. phrases
Murder	30	1,560	149	18
Land Dispute	25	2,461	196	22
Robbery	21	1,907	162	17
Corruption	17	2,542	197	16
All	93	2,023	173	18

ROUGE-L, which measure overlap based on the words and phrases present in the summaries.

To compute these lexical metrics, we generate summaries from sentences containing the relevant intent-related terms. We then compare the performance of all metrics against the ratings produced by the LLM-as-Judge framework, which serves as a proxy for human evaluation in our experiments.

This evaluation setup follows the methodology used in prior studies, where automated metrics were compared against judgment-based scoring approaches.

IV. EXPERIMENTAL SETUP

A. Dataset

We use the Indian-Data corpus released by [9]: 93 Indian Supreme Court / High Court judgments across four categories with manually annotated intent phrases. Statistics are summarised in Table I.

B. Summarization

We generate extractive summaries using the BERT-based clustering method of Miller [8]: sentences are encoded with all-MiniLM-L6-v2, clustered with k -means, and the sentence closest to each cluster centroid is selected. The number of clusters is set to 30% of the source sentence count, matching the ratio used by [9].

C. SIM Threshold Selection

We try out values for τ from 0.40 to 0.80 and pick the one that gives us the best mean F1 score. We want to find the spot where τ is between 0.45 and 0.70 so that our metric is not too perfect, counting every match when τ is low, or too empty with no matches when τ is high. We look at the results of trying out all these τ values in Figure 1. We choose $\tau = 0.45$ because it gives us a precision of 0.585, a mean recall of 0.923, and a mean F1 score of 0.708. We do not choose $\tau = 0.40$ because the precision is too low, below 0.67, and the recall is already very high, which means we are being too loose with our matching.

D. Implementation

All experiments run on a single Colab T4 GPU. SIM uses all-mpnet-base-v2 (438MB). LLM-as-Judge uses Qwen2.5-1.5B-Instruct (3GB) loaded locally; no API access required. Total runtime for the 93-document evaluation is approximately 50 minutes. Code is released in a single Jupyter notebook for reproducibility.

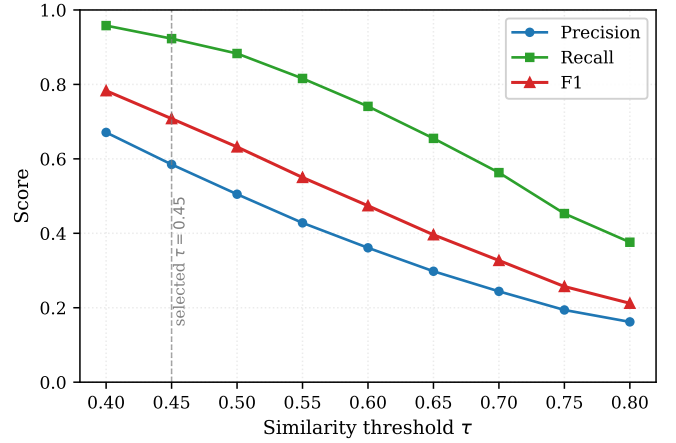


Fig. 1. SIM precision, recall, and F1 as a function of the cosine threshold τ . Selected $\tau=0.45$ (dashed line) maximises F1 in the operating range $\tau \in [0.45, 0.70]$.

TABLE II
MEAN METRIC SCORES ON INDIAN-DATA ($n=93$).

Metric	Mean
BLEU	0.190
ROUGE-L	0.259
Original Intent F1 [9]	0.118
SIM F1 (ours)	0.708
LLM-as-Judge (1-5)	4.28

V. RESULTS

A. Metric Averages

Table II reports the mean of each metric across the 93 documents at $\tau=0.45$ for SIM.

The Original Intent Metric scores low (0.118) because BERT-based extractive summarization rarely copies annotated intent phrases verbatim. SIM recovers most of this “missing” intent through semantic matching, scoring 0.708, an absolute gain of 0.59.

B. Correlation with LLM-as-Judge

Table III reports the Spearman rank correlation between each automated metric and the LLM-as-Judge score. SIM is the only metric showing a significant positive correlation ($\rho=+0.255$, $p=0.014$). The original Intent Metric correlates significantly *negatively* with the LLM-Judge ($\rho=-0.212$, $p=0.041$); BLEU and ROUGE-L show weak negative correlations.

Figure 2 visualises the full metric-vs-metric correlation matrix. BLEU, ROUGE-L, and Original Intent are tightly correlated with each other ($\rho>0.83$), reflecting that all three reward verbatim overlap with the source. SIM and LLM-Judge form a separate cluster: each correlates weakly negatively with the lexical group, and positively with each other. This suggests SIM and LLM-Judge measure a distinct property—semantic intent preservation—that is not captured by the lexical family.

TABLE III
SPEARMAN RANK CORRELATION BETWEEN AUTOMATED METRICS AND LLM-AS-JUDGE SCORES ON INDIAN-DATA ($n=93$). * INDICATES $p < 0.05$.

Metric	Spearman ρ	p -value
BLEU	-0.173	0.097
ROUGE-L	-0.164	0.116
Original Intent F1	-0.212	0.041*
SIM F1 (ours)	+0.255	0.014*

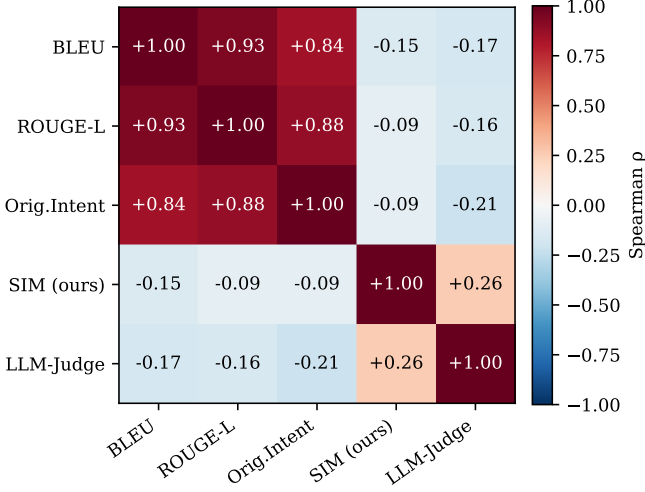


Fig. 2. Spearman rank correlation between all metric pairs on Indian-Data. Lexical metrics (BLEU, ROUGE-L, Original Intent) form one cluster; SIM and LLM-Judge form another.

C. Per-Category Analysis

Figure 3 compares per-category F1 between the original metric and SIM. The absolute gain is largest on Land Dispute (+0.63) and Murder (+0.61), and smallest on Robbery (+0.50). More informatively, Table IV reports the per-category Spearman correlation of SIM with LLM-Judge.

SIM is significantly positive on Corruption and Land Dispute, near zero on Murder, and weakly negative on Robbery. We interpret this as a consequence of intent phrase character: Corruption and Land Dispute phrases tend to be more abstract and legalistic (“*accepted illegal gratification*”, “*adverse possession*”), where summaries are likely to paraphrase. Murder and Robbery phrases are often concrete and physical (“*stabbed with knife*”, “*robbed articles*”), where exact-match and semantic-match agree more closely, leaving SIM with less “room to add value”.

D. SIM vs. LLM-Judge Scatter

Figure 4 shows the per-document scatter. Documents receiving an LLM-Judge score of 5 cluster at higher SIM F1 than those receiving 3 or 4, consistent with the positive Spearman correlation. The relationship is monotonic but noisy, reflecting the discreteness of the LLM-Judge scale and the heterogeneity of legal documents within each category.

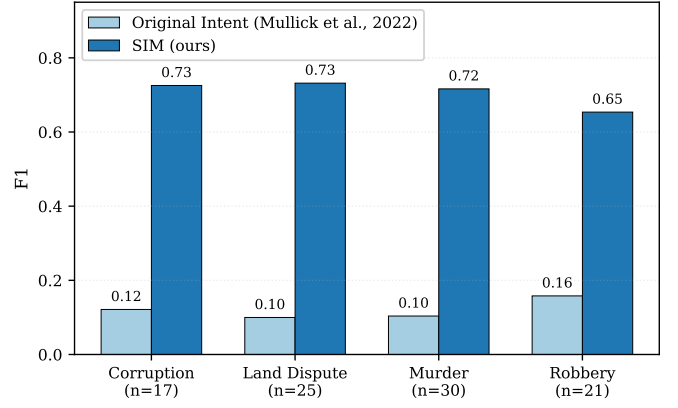


Fig. 3. Per-category mean F1: Original Intent vs. SIM. SIM consistently recovers 0.5–0.6 in absolute F1 across all categories.

TABLE IV
PER-CATEGORY SPEARMAN CORRELATION BETWEEN SIM AND LLM-JUDGE. SIM IS SIGNIFICANTLY POSITIVE ON CORRUPTION AND LAND DISPUTE. * INDICATES $p < 0.05$.

Category (n)	SIM ρ	p -value
Corruption (17)	+0.488	0.047*
Land Dispute (25)	+0.464	0.020*
Murder (30)	+0.200	0.289
Robbery (21)	-0.236	0.303

VI. DISCUSSION

A. Why Does the Original Intent Metric Correlate Negatively with LLM-Judge?

This is the most surprising finding. We attribute it to the behaviour of the BERT extractive summarizer used in our experiments: it tends to select long, procedurally-rich sentences (e.g., headers, citations, court names) that happen to contain verbatim intent phrases as fragments. The original metric scores these summaries high because the substring is present, but the surrounding context is procedural noise that the LLM-Judge penalises. SIM, by contrast, requires *semantic* similarity in a length-matched window, which discounts surrounding noise. Lexical metrics inherit the same bias as the original metric: they reward overlap with sentences that are intent-bearing-but-procedural, which the LLM does not view as good summaries.

B. Where SIM Helps Most

SIM’s added value is concentrated on cases where intent is encoded in legalistic abstractions rather than physical descriptors. This pattern aligns with intuition about the corpus: legal disputes over corruption or property rights are framed in interpretive language that admits many surface forms, whereas violent-crime descriptions tend to use a more standardised concrete vocabulary.

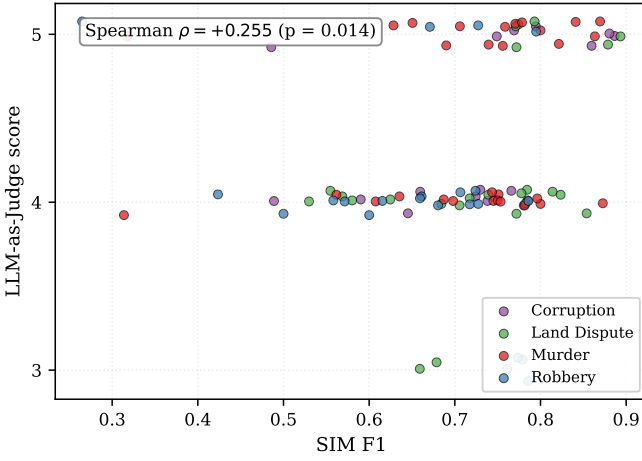


Fig. 4. Per-document SIM F1 against LLM-as-Judge score, coloured by category. Vertical jitter is added to LLM scores (integers 3–5) for visibility.

C. Why a Sentence-Tuned Encoder Rather than Legal-BERT

In a pilot, we compared mean-pooled `nlpaueb/legal-bert-base-uncased` embeddings against `all-mpnet-base-v2`. Raw Legal-BERT mean-pooled embeddings of legal phrases consistently fell within a narrow cosine band (~ 0.6 – 0.7) regardless of whether the pair was a true match or merely topical, giving a poor decision boundary. MPNet, contrastively pretrained on sentence-pair similarity, separates matches from non-matches with a wider gap. We therefore use MPNet despite Legal-BERT’s domain familiarity.

VII. LIMITATIONS

Reference signal is an LLM, not human ratings. The original [9] framework used crowd-sourced human ratings on the Appen platform. Our reference signal is a small open-source LLM (Qwen2.5-1.5B). LLM-as-Judge scores can systematically diverge from expert human judgement [18], and our positive correlation between SIM and LLM-Judge could in part reflect shared inductive biases between MPNet and Qwen rather than alignment with human preference. Validation against expert human annotation, in particular by lawyers, is left to future work.

Single corpus. We evaluate only on Indian-Data (93 documents). Generalisation to the Australian-Data corpus released by [9], or to legal corpora from other jurisdictions, has not been verified.

Single summarizer. All experiments use the BERT-extractive summarizer of Miller [8]. The relative ranking of metrics may differ for graphical-CRF [15], abstractive [1], or LLM-based summarizers.

Threshold selection. The SIM threshold $\tau=0.45$ is selected on the same 93 documents on which we report results, raising the possibility of mild over-fitting. A held-out tuning set would be preferable; we did not split the data given its small size.

Statistical power. With $n=93$, the per-corpus Spearman correlations are significant at $p<0.05$ but the per-category

breakdown ($n=17$ – 30) has low power, particularly for Murder and Robbery.

VIII. CONCLUSION

We extended the intent-aware evaluation framework of Mullick et al. [9] along two axes: (1) replacing exact-substring matching with windowed cosine similarity over MPNet embeddings (SIM), and (2) integrating an LLM-as-Judge reference signal. On 93 Indian legal documents, SIM is the only automated metric that correlates positively and significantly with LLM-Judge scores ($\rho=+0.255$, $p=0.014$); BLEU, ROUGE-L, and the original Intent Metric correlate weakly negatively. The improvement is concentrated on Corruption and Land Dispute cases, where intent phrases admit more paraphrase. SIM is free, reproducible, and computable in ~ 10 minutes on a free-tier GPU, suggesting it is suitable as a deployment metric where access to expert human raters is impractical.

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